Priyanshu-sharma CU2402250980
Full-Stack TEST.

① Node.js - The Event loop.

The Console Event loop →

A
D
C
B

② Node.js – Modules

(Calc.js)

```
Function add (a,b){ return
  a+b;}
Function Subtraction(a,b){....
  return  a-b;}
Module.exports = { add , Subtract};
```

(App.js)

```
Const {add , Subtraction} =
  require("./calc");
Console.log(add(4,5));
Console.log(Subtract(4,5));
```

③ Express.js – Middleware + Routing.

```
Function
requireAuth (req , res , next) {
  !== "secret123")
      return
res.status(401).json({error
  :"unauthorized"});
    next();
}
app.get("/api/books",
(req , res) => res.json(books));
app.post("/api/books",
requireAuth , (req , res => {
res.status(201).json(req.body);
```

④ Express.js - Building a REST API

```
app.get("/api/books",
  (req, res) => res.status(200)
  .json(Books));
app.post("/api/books",
  (req, res) => {
    const book = {id: books.length + 1 - - req.body};
    books.push(book);
  res(status(201).json(book);
  });
  res.status(200).json({message:"Deleted"});
```

⑤ Express.js - Debugging + Error Handling.

```
app.get("/api/books/:id",
  (req, res) => {
      const book =
  book.find(b => b.id ==
    req.params.id);
      IF(!book)
        return
  res.status(404).json({error:"Book not Found"});
    res.json(book);
  });
```

⑥ APIs - Consuming a REST API in React

```
use Effect(() => {
Fetch("http://localhost:
  5000/api/books")
    .then(r => r.json())
    .then(data => setBooks(data))
    .catch(() => set Error("Failed to load"));
  }, []);
```

⑦ APIs-CORS

```
Const Cors =
    require("Cors");
    app.use(Cors({
        Orisin
    "http://localhost:3000"
    }));
```